

LIBIN SAMKUTTY

Senior QA Automation Engineer | Vadodara (Open to Relocate) | samlibin2001@gmail.com | linkedin.com/in/libin-samkutty | libin-samkutty.github.io

PROFESSIONAL SUMMARY

Senior QA Automation Engineer with 4.5+ years owning test automation architecture, framework design, and quality strategy across a cloud-native, AI-powered digital health platform serving 50,000+ monthly users across 6 countries and 6+ concurrent product lines. Designed and delivered Playwright (300+ scenarios), Botium conversational AI (140+ flows), and Pytest+Pact API automation suites integrated into GitHub Actions and CircleCI CI/CD pipelines; co-designed a real-time ML-monitoring pipeline (AWS MSK/Kafka, Lambda, LangSmith) that caught a clinical-safety gap invisible to offline testing; and published an independent open-source PyPI testing tool with 1,800+ downloads.

CORE COMPETENCIES

Test Automation Architecture, Playwright, Pytest, Robot Framework, Botium, Selenium, Page Object Model (POM), BDD/Gherkin, CI/CD Integration, GitHub Actions, CircleCI, Jenkins, Docker, API Testing, Contract Testing (Pact), REST API, Schema Validation, OAuth/JWT, OWASP ZAP, DAST, Security Testing, AI Quality Engineering, LLM Evaluation, Non-Deterministic Testing, RAG Testing, Performance Testing, JMeter, Shift-Left Testing, SQL, Data Validation, Python, JavaScript, TypeScript, AWS, GCP, Agile/Scrum, JIRA, Defect Management, Release Quality Gates

WORK EXPERIENCE

Senior QA Automation Engineer | Jeavio India Pvt. Ltd. (Client: Nivi)

Jun 2026 - Present

- Promoted to Senior QA Automation Engineer in recognition of scope carried as sole QA across 6+ AI-powered product lines, including ML-monitoring infrastructure ownership, a multi-phase AI evaluation framework, and direct client-facing engagement with Nivi's CTO and CEO.
- Leading migration of floating RAGAS/G-Eval CI thresholds to a custom binary PASS/FAIL judge with explicit per-dimension criteria checklists, replacing unprincipled score gates with defensible, calibrated failure boundaries.
- Co-designed a real-time evaluation pipeline (**AWS MSK/Kafka, Lambda, LangSmith**) that, within 2 weeks of go-live, surfaced a clinical-safety gap invisible to golden-dataset, routing, and manual-review testing, restoring Clinical Safety PASS rate to **94%** on the India Maternal Health sample after driving retrieval, prompt, and dataset fixes.

QA Engineer | Jeavio India Pvt. Ltd. (Client: Nivi)

Jun 2023 - May 2026

- Architected and led migration of the UI automation suite from Robot Framework (BDD/Gherkin/pabot) to **Playwright** with Page Object Model design and native 6-way CI sharding across GitHub Actions runners, cutting smoke-suite runtime **~61%** (~28 to ~11 min) and full regression-suite runtime **~65%** (~65 to ~22-24 min), growing the suite to 300+ end-to-end and integration scenarios; converted ~25-30% of browser-unnecessary scenarios to pure API tests, further reducing regression cycle time.
- Implemented consumer-driven **contract testing (Pact)** between the Journey Engine and NLP Classification Service, wiring a **can-i-deploy** gate to a self-hosted Pact Broker and an OpenAPI YAML schema-diff engine for breaking-change detection, catching a 1,800+ event-row misclassification bug in provider verification before it reached staging, converting a class of silent integration break that once took 2 days to diagnose manually into a build-time failure.
- Designed and delivered a **Botium** -based chatbot automation framework covering 140+ conversational flow scenarios (multi-turn journeys, intent routing, adversarial inputs) and integrated it into **GitHub Actions/CircleCI** pipelines; combined with Playwright automation, reduced manual regression effort from 5 to 2 people-days per release cycle (**60% reduction**) and drove test coverage from 0% to 75%+ across 440+ scenarios.
- Established the platform's first structured **DAST security-testing** program (OWASP ZAP, full/active scan, JWT-authenticated), surfacing 2 High-severity vulnerabilities (stored XSS, missing CSRF token) on the first scan, both fixed within the sprint, and delivering per-sprint scan reports as **ISO 27001** compliance evidence.

Associate QA Engineer | Jeavio India Pvt. Ltd. (Client: Nivi)

May 2022 - May 2023

- Developed the platform's first structured QA process from zero, test case templates, coverage matrices, severity classification, defect governance workflows, adopted across 3+ engineering teams, reducing defect escape rate ~15%; created **Pytest/Requests** API automation covering 30+ REST endpoints across 5 microservices and the original Robot Framework (BDD/Gherkin) UI suite, cutting CI smoke-suite runtime 62 to 17 minutes via pabot **--testlevelsplitt** parallelization.
- Surfaced 20+ critical/high-severity defects pre-release through systematic negative testing, boundary value analysis, and SQL-based validation of 6+ ETL pipeline outputs, catching 15+ data anomalies before they could corrupt analytics dashboards serving 50,000+ monthly users.

Project Intern | Jeavio India Pvt. Ltd.

Jan 2022 - Apr 2022

- Executed manual functional and integration testing for NLP intent classification and NER-based conversational flows; supported API validation using Postman and contributed to early-stage QA process documentation and test environment configuration.

SKILLS

Test Automation Frameworks: Playwright (POM, native sharding, visual regression), Pytest (fixtures, parametrize, BDD-style), Robot Framework (Gherkin/BDD, pabot), Botium (conversational AI), Selenium, Postman/Newman, JMeter (baselining, threshold gates, InfluxDB/Grafana trend monitoring), pytest-benchmark

AI & LLM Testing: DeepEval (G-Eval), RAGAS (Faithfulness, Context Precision/Recall), LLM-as-a-Judge design and calibration, golden dataset curation, NLP intent classification testing, semantic cache validation (Valkey/precision-recall harness), guardrail/prompt tuning, non-deterministic output testing

API & Contract Testing: REST API automation, Pact (consumer-driven contracts, Pact Broker, can-i-deploy gating), OpenAPI/YAML schema-diff, schema validation, OAuth/JWT authentication, negative testing, data-driven execution

CI/CD & DevOps: GitHub Actions, CircleCI, Jenkins, Docker, parallel/sharded execution, Allure reporting, release quality gates, JIRA (Xray), Git

Production AI Monitoring: AWS MSK (Kafka), AWS Lambda (Event Source Mapping, DLQ), LangSmith (trace visibility, annotation queues), Prometheus, Grafana

Security & Compliance: OWASP ZAP (DAST, full/active scan, JWT-authenticated), XSS/CSRF vulnerability triage, ISO 27001 compliance evidence

Languages & Data: Python, JavaScript, TypeScript, SQL (JOINS, window functions, CTEs, geospatial), BigQuery, PostgreSQL

Cloud & AI Tools: AWS (EC2, Lambda, MSK, RDS), GCP (BigQuery, Vertex AI), Claude API, Gemini API, OpenAI API, GitHub Copilot, ChatGPT

PROJECTS

api-test-data-generator | Python, Faker, Pytest, JSON/YAML | pip install api-test-data-generator

- Designed and published a **schema-driven Python package** for API test data generation with native Pytest fixture integration, reaching **1,824 total PyPI downloads** (391 in the last 30 days as of June 2026) with sustained usage well beyond the launch spike, demonstrating independent software engineering capability.
- Engineered **JSON/YAML declarative schema definitions** with field-level type and constraint support, Faker-backed realistic data generation, and JSON/CSV export, with Pytest fixture integration as a first-class feature enabling drop-in adoption into existing API automation workflows; MIT licensed and publicly available.

EDUCATION

Bachelor of Engineering (B.E.), Computer Engineering | S.V.I.T., Vasad - GTU

2022 | CGPA: 9.35 / 10